THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL2108 PRINCIPLES OF CIVIL ENGINEERING MANAGEMENT

May 16, 2016 14:30 pm – 17:30 pm (3 hours)

Answer **ANY FOUR** questions

All questions carry equal marks

Use of Calculators

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

1. To cope with the increasing demand for student accommodation, the university is planning to build a number of new residential halls in the coming years.

   a) Advise the university on the suitability of open tendering and negotiation in the context of residential halls construction. [15 marks]

   b) Describe what is serial tendering, and discuss whether this tendering approach would benefit the university if the planned residential halls are located on different sites and will be built at different timeframe. [10 marks]

2. a) Calculate the internal rate of return of a power generation plant based on the following parameters:

   Construction cost = $20 billion
   Operation period = 20 years
   Electricity generated = $4 billion per annum in the first ten years
   and subsequently at $6 billion per annum
   Maintenance and operating costs = $1.6 billion per annum
   Replace generator components = $3 billion every 6 years
   Reversion value = $4 billion [18 marks]

   b) Briefly describe the concept of residual method and explain whether it is an appropriate approach for evaluating the financial feasibility of this scheme. [7 marks]

3. a) Outline the key elements making up a unit rate and explain what should be considered from the detailed cost estimation sense in order to arrive at a reliable value for each of the identified cost elements. [15 marks]

   b) Describe three common factors related to raw materials which should be built into the unit rate calculation of cast in-situ concrete. [6 marks]

   c) Justify whether the unit rate of a 500 × 500 × 4,000 mm cast in-situ concrete column should be the same as that of a 1,250 × 1,250 × 640 mm cast in-situ foundation. [4 marks]

4. a) The key role of a project manager is to ensure a project would meet or even outperform the time, cost and quality requirements. Identify the tasks a project manager should perform during the pre and post contract stages in order to minimise the time, cost and quality risks of the client. [13 marks]

   b) Describe with reasons the information and records to be maintained by contractor's site personnel while the work is in progress to support various management decisions. [12 marks]

5. The following is the price breakdown of the lowest tender received. After careful examination, it is found that the tenderer has made an arithmetical mistake and the correct total should be HK$124,636,062 instead.

	HK$
Preliminaries	15,319,140
Substructure	11,350,420
Superstructure	90,196,686
External Works	1,869,816
Prime Cost and Provisional Sums	5,900,000
Total carried to form of tender:	130,636,062

a) Under the Code of Procedures for Single Stage Selective Tendering, discuss what options are available to the client and tenderer when tender errors are discovered in a lump sum fixed price contract. [6 marks]

b) Calculate the percentage error (and show clearly whether it is a plus or minus percentage) based on the price breakdown shown above. [8 marks]

c) Based on (b) above, explain clearly the course of action to be taken by the client before the lump sum fixed price contract is awarded and describe the consequence at the post contract stage using the result derived in (b)? [6 marks]

d) If the tender price is used as a basis for negotiation, discuss whether the course of action should be the same as that in (c). [5 marks]

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